

Principles of Applied Engineering

TEKS §127.781 • CTE Level 1 • Grade 9 • Mr. Gavaldon • Da Vinci School

Week 2

MONDAY

Day 6 - Build-a-Car: Plan & Design

Aug 17

OBJECTIVE

Students will plan a small rolling car from minimal materials, finishing a design/planning sheet before building.

TEKS

§127.781(d)(2)(G) engineering design process — planning; (d)(3)(C) sketching the design

ELPS / EB SUPPORT

Planning sheet gives frames + a labeled diagram (VC, CD). Mixed teams with builder / recorder / tester roles (PN).

AGENDA • 45 MIN

5' Bell-work: what makes a car roll far?

10' Design brief + constraints

8' Recap design process - plan first

17' Sketch design + materials list on planning sheet

5' Predict distance + exit

SLIDES / ACTIVITY

Design a Car That Rolls

NOTEBOOK

Glue in the car planning sheet. Sketch your design and label the body, axles, and wheels.

TURN IN

Car Design - Planning Sheet

TUESDAY

Day 7 - Build Day

Aug 18

OBJECTIVE

Students will build their prototype car from their approved plan, testing and adjusting as they go.

TEKS

§127.781(d)(2)(G) build the prototype (design process); (d)(4) safe, responsible use of tools & materials

ELPS / EB SUPPORT

Roles keep every student hands-on (CD). Teacher models assembly at the front before teams start (VC, RS).

AGENDA • 45 MIN

5' Bell-work: name one build risk to avoid

8' Teacher models axle/wheel assembly

27' Teams build from approved plans (table roll-tests)

5' Park it + note one fix for tomorrow

SLIDES / ACTIVITY

Build Your Prototype

NOTEBOOK

Log your build: what went well, what didn't, and the one fix you'll make on test day.

TURN IN

— (notebook / no turn-in)

WEDNESDAY

Day 8 - Test, Measure, Improve

Aug 19

OBJECTIVE

Students will test their car, measure distance, make one controlled improvement, and reflect using design-process language.

TEKS

§127.781(d)(2)(G) test & evaluate (design process); (d)(3)(D) present results with visualization

ELPS / EB SUPPORT

Data sheet is mostly numbers + one sentence (low language load, CD). Pair-share the reflection first (WT, PN).

AGENDA • 45 MIN

5' Bell-work: what is a fair test?

8' How we test + record

20' Ramp test: 2 trials + one improvement

7' Data sheet + reflect

5' Exit ticket

SLIDES / ACTIVITY

Test & Improve

NOTEBOOK

Record both trial distances and the single change you made. Was your prediction right?

TURN IN

Car Test & Improve - Data Sheet

THURSDAY**Day 9 - The Real Tools (video + emails)**

Aug 20

OBJECTIVE

Students will describe what Fusion 360 is and why we use it, and confirm a working email so accounts can be created next class.

TEKS

§127.781(d)(3)(C) computer-aided design & drafting (CADD); (d)(1)(A) professional online conduct

ELPS / EB SUPPORT

Video has a short guided sheet with stems (RS, CD). 'Show me your email' is a one-on-one check - low stakes (WT).

AGENDA • 45 MIN**5'** Bell-work: what is CAD?**12'** What is Fusion 360? + watch intro**8'** Where this leads: certification path**15'** Email check as a station — everyone else does the Fusion-preview notebook entry**5'** Info card + exit**SLIDES / ACTIVITY**

From Paper to Real CAD

NOTEBOOK

Note 3 things Fusion can do, and one thing you want to build in it this year.

TURN IN

Account Setup - Email Check (info card)

FRIDAY**Day 10 - Create Account & Launch Fusion**

Aug 21

OBJECTIVE

Students will create an Autodesk education account and launch Fusion 360, locating key tools and saving a first file.

TEKS

§127.781(d)(3)(C) set up & navigate CAD (CADD); (d)(1)(B) collaborate & support peers

ELPS / EB SUPPORT

Steps are numbered + screenshotted in Classroom (VC). Finishers pair with EB students to walk them through sign-in (PN).

AGENDA • 45 MIN**5'** Bell-work: why a school email for this?**8'** Account steps (invites sent Day 9)**25'** Create/verify account -> launch Fusion; if stuck, start the Fusion-preview fallback**7'** Checklist + help a neighbor**SLIDES / ACTIVITY**

Create Your Account & Open Fusion

NOTEBOOK

Paste your Fusion account email (not password). Note where the Sketch button and timeline are.

TURN IN

Fusion 360 - First Look Checklist